

DSA / Algorithms (Abdul Bari) - Day 14

45-day plan mapped to the Algorithms playlist

Merge Sort

Two-way merge sort, the algorithm and its analysis.

Watch these Abdul Bari videos

- > 2.7.1 Two Way MergeSort - Iterative method
- > 2.7.2. Merge Sort Algorithm
- > 2.7.3 MergeSort in-depth Analysis

Key concepts

- Divide, sort halves, merge.
- Stable, $O(n \log n)$, $O(n)$ extra space.
- Merge step combines two sorted runs.

Key idea

`merge()` walks both halves picking the smaller front element.

Tasks (1.5h)

- ☐ Watch the 3 videos
- ☐ Implement merge sort
- ☐ Analyse its complexity

Playlist: youtube.com/playlist?list=PLDN4rrl48XKpZkf03iYFl-029szjTrs_0